WHILE LOOP

ITERATION STATEMENT

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

while Loop

WHILE LOOP

```
while (condition) {
    statement
}
```

The while loop **repeats** the statement while the condition is **true**

If the condition is **no longer true**, the loop ends

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

n--

decreased the value of n by one

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (10 > 0){

    cout<< 10 << ", ";

    9;

}
```

output: 10,

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (9 > 0){

    cout<< 9 << ", ";

    8;

}
```

output: **10, 9,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (8 > 0){

    cout<< 8 << ", ";

    7;

}
```

output: **10, 9, 8,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (7 > 0){

    cout<< 7 << ", ";

    6;

}
```

output: **10, 9, 8, 7,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (6 > 0){

    cout<< 6 << ", ";

    5;

}
```

output: **10, 9, 8, 7, 6,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (5 > 0){

    cout<< 5 << ", ";

    4;

}
```

output: **10, 9, 8, 7, 6, 5,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (4 > 0){

    cout<< 4 << ", ";

    3;

}
```

output: **10, 9, 8, 7, 6, 5, 4,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (3 > 0){

    cout<< 3 << ", ";

    2;

}
```

output: **10, 9, 8, 7, 6, 5, 4, 3,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (2 > 0){

    cout<< 2 << ", ";

    1;

}
```

output: **10, 9, 8, 7, 6, 5, 4, 3, 2,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (1 > 0){

    cout<< 1 << ", ";

    0;

}
```

output: **10, 9, 8, 7, 6, 5, 4, 3, 2, 1,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (0 > 0){
    ----------
    ----------
}
```

output: **10, 9, 8, 7, 6, 5, 4, 3, 2, 1,**

```
// countdown
#include <iostream>
using namespace std;
int main( ) {
    int n = 10;

    while (n > 0){
        cout<< n << ", ";
        n--;
    }

    cout<<"liftoff!";

    return 0;
}
```

```
while (0 > 0){
    ----------------------
    ----------------------
}
```

output: **10, 9, 8, 7, 6, 5, 4, 3, 2, 1, liftoff!**

```
char yn = 'y';
cout<<"Do you wish to continue?<y/n>: ";
cin>>yn;

while(yn!='y' || yn!='n') {
    cout<<"Please enter 'y' or 'n' only: ";
    cin>>yn;
}

cout<<"- end of while loop - ";
```

LABORATORY